Final Exam, Revision 3:27pm

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1. Consider the IBVP:

$$u_{tt} - u_{xx} = 0,\ u(0,t) = 0,\ u(x,0) = 0, u_t(x,0) = f(x), \tag{1}$$

where $x > 0$, $t > 0$, $f(x) = 1$ for $0 \leq x \leq 1$ and $f(x) = 0$ for $x > 1$.

Proof (Hernandez). When $x > 1$, we check that $u(x,t) = 0$ is a solution.

When $0 \leq x \leq 1$, we obtain a solution by d'Alembert's formula given below:

$$u(x,t) = \frac{0+0}{2} + \frac{1}{2 \cdot 1} \int_{x-t}^{x+t} f(\xi)\ d\xi.$$

□

2.

$$u_t - u_{xx} = \frac{1}{x^2+1},\ \ -\infty < x < \infty, t > 0, u(x,0) = 0. \tag{2}$$

Proof (Hernandez). According to Duhamel's principle, the solution is given in terms of the heat kernel G as shown below:

$$u(x,t) = \int_0^t \int_{-\infty}^{\infty} \frac{G(x-y, t-\tau)}{y^2+1}\ dyd\tau = \int_0^t \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi k(t-\tau)}} e^{-\frac{(x-y)^2}{4k(t-\tau)}} \frac{1}{y^2+1}\ dyd\tau.$$

□

3.

$$u_{tt} - u_{xx} = x,\ u(x,0) = 0,\ u_t(x,0) = 1, u_x(0,t) = u_x(\pi,t) = 0, \tag{3}$$

where $0 < x < \pi$, $t > 0$.

Proof (Hernandez). The solution is given by a superposition $u = v + w$ solving:

$$v_{tt} - v_{xx} = 0,\ v(x,0) = 0,\ v_t(x,0) = 1, v_x(0,t) = v_x(\pi,t) = 0,$$
$$w_{tt} - w_{xx} = x,\ w(x,0) = 0,\ w_t(x,0) = 0, w_x(0,t) = w_x(\pi,t) = 0,$$

Clearly, $v(x,t) = t$. For w, we obtain the following eigenfunctions:

$$-\varphi_n''(x) = \lambda^2 \varphi_n(x), \qquad \varphi_(0) = \varphi(\pi) = 0,$$
$$\varphi_n(x) = \sin(\lambda x), \qquad \lambda = \frac{n\pi \cdot 1}{\pi} = n,$$

which leads to the following solution:

$$w(x,t) = \sum_{n=1}^{\infty} \left(\frac{1}{n} \int_0^t f_n(s) \cos(ns)\ ds \right) \sin(nt) \sin(nx),$$

where $f_n(s)$ is the Fourier coefficient of x as shown below:

$$x = \sum_{n=1}^{\infty} f_n(t) \sin(nx).$$

□

4.

$$u_{tt}+u_{xxxx}=0, u(0,t)=u_{xx}(0,t)=0, u_x(1,t)=u_{xxx}(1,t)=0, u(x,0)=1, u_t(x,0)=\sin(3\pi x/2), \tag{4}$$

where $0<x<1$, $t>0$.

Proof (Hernandez). We obtain the following solution:

$$u(x,t)=\sum_{n=0}^{\infty}\frac{1}{n}\int_{-\infty}^{\infty}\frac{\sin(3\pi w/2)}{w^2}\sin(3\pi wt/2)e^{wtx}\ dw.$$

□

5.

$$u_{tt}=u_{xx}-2u,\ u_x(0,t)=0, u(1,t)=0, u(x,0)=1, u_t(x,0)=0, \tag{5}$$

where $0<x<1$, $t>0$.

Proof (Hernandez). Suppose $u(x,t)=\varphi(x)g(t)$. Then, we obtain the following ODEs:

$$\ddot{g}=(\lambda^2-2)g \qquad \varphi''=-\lambda^2\varphi, u_x(0,t)=\varphi'(0)=0$$
$$g_n(t)=a\cos(t\sqrt{\lambda_n^2+2})+b\sin(t\sqrt{\lambda_n^2+2}), \qquad \varphi_n(x)=c\cos(\lambda_n x).$$

Applying $u(1,t)=c\cos(\lambda_n\cdot 1)=0$ yields $\lambda_n=\arccos(\cos(\lambda_n))=\arccos(0/c)=\frac{\pi(2n-1)}{2}$, and superposition yields:

$$u(x,t)=\sum_{n=1}^{\infty}\cos(\lambda_n x)\left[A_n\cos(t\sqrt{\lambda_n^2+2})+B_n\sin(t\sqrt{\lambda_n^2+2})\right]$$

Applying $u_t(x,0)=\sum_{n=1}^{\infty}\cos(\lambda_n x)\left[-A_n\sin(0\sqrt{\lambda_n^2+2})+B_n\cos(0\sqrt{\lambda_n^2+2})\right]\sqrt{\lambda_n^2+2}=0$, equivalently $u_t(x,0)=\sum_{n=1}^{\infty}\cos(\lambda_n x)B_n\sqrt{\lambda_n^2+2}=0$ implies $B_n=0$, and A_n can be computed as the following Fourier coefficient:

$$A_n=2\int_0^1\cos(\lambda_n x)\ dx=\frac{2}{\lambda_n}\sin(\lambda_n).$$

Thus, we obtain the following solution:

$$u(x,t)=\sum_{n=1}^{\infty}\sin(\lambda_n x)\cos(\lambda_n x)\cos\left(t\sqrt{\lambda_n^2+2}\right), \qquad \lambda_n=\pi\left(n-\frac{1}{2}\right).$$

□

6.

$$\frac{1}{2x}u_x+u_t-u=0,\ -\infty<x<\infty, t>0, u(x,0)=x^2 \tag{6}$$

Proof (Hernandez). We rewrite this equation as follows:

$$u_t + c(x,t)u_x = f(x,t,u),$$

where $c(x,t) = c(x) = \frac{1}{2x}$ and $f(x,t,u) = u$, and transform to characteristic coordinates $U(\xi,\tau)$ where $\tau = t$ and $\xi = \phi(x,t) = C = x^2 - t$ solves the following ODE:

$$x^2 = \int 2x \cdot \frac{dx}{dt}\, dt = \int 2x \cdot \frac{1}{2x}\, dt = t + C.$$

Then, we obtain the following solution:

$$\begin{aligned} [-U_\xi + U_\tau] + \frac{1}{2x}\left[2x \cdot U_\xi\right] = U_\tau = U \qquad & \text{by substitution and chain rule} \\ U(\xi,\tau) = e^\tau \xi \qquad & \text{by anti-differentiation} \\ u(x,t) = e^t(x^2 - t) \qquad & \text{re-coordinatization.} \end{aligned}$$

□

7.

$$(x+t)u_x + tu_t + u = 0, \;\; -\infty < x < \infty, \;\; t > 1, \;\; u(x,1) = x. \tag{7}$$

Proof (Hernandez). We rewrite this equation as follows:

$$u_t + c(x,t)u_x = f(x,t,u),$$

where $c(x,t) = \frac{x+t}{t}$ and $f(x,t,u) = -\frac{u}{t}$, and transform to characteristic coordinates $U(\xi,\tau)$ where $\tau = t$ and $\xi = \phi(x,t) = C = x/t - \ln(t)$ solves the following ODE:

$$\begin{aligned} \frac{dx}{dt} = c(x,t) = \frac{x}{t} + \frac{t}{t} \\ \frac{1}{t}\frac{dx}{dt} - \frac{x}{t^2} = \left[\frac{dx}{dt} - \frac{x}{t}\right] e^{\int -\frac{1}{t}} = \left[\frac{1}{1}\right] e^{\int -\frac{1}{t}} = 1 \cdot \frac{1}{t} \\ x/t = \int \frac{d\left(\frac{x}{t}\right)}{dt} = \int \frac{1}{t}\, dt = \ln(t) + C \end{aligned}$$

Then, we obtain the following solution:

$$\begin{aligned} \left[U_\tau - \frac{x+t}{t^2}U_\xi\right] + \frac{x+t}{t}\left[\frac{1}{t}U_\xi\right] = U_\tau = -\frac{1}{t}U \qquad & \text{by substitution and chain rule} \\ U(\xi,\tau) = \frac{\xi}{\tau} \qquad & \text{by anti-differentiation} \\ u(x,t) = \frac{x/t - \ln(t)}{t} = \frac{x - \ln(t)t}{t^2} \qquad & \text{by re-coordinatization.} \end{aligned}$$

□

8. Consider the following IBVP in the unit 3-d sphere:

$$u_{tt} = \Delta u, 0 < r < 1, u|_{r=1} = 0, \;\; u|_{t=0} = r, u_t|_{t=0} = 1. \tag{8}$$

Proof (Hernandez). Our equation is $u_{tt} = \Delta u = u_{rr} + \frac{2}{r}u_r$, giving a solution of the following form:

$$u(r,t) = \sum_{n=1}^{\infty} c_n(t)y_n(r) = \sum_{n=1}^{\infty} \alpha_n \frac{\sin(\pi n t)}{\pi n} y_n(r), \qquad y_n(r) = \frac{\sin(n\pi r)}{r},$$

where α_n is given as the following Fourier coefficient:

$$\alpha_n = \frac{1}{1/2}\int_0^1 1 \cdot \sin(\pi n r) r \; dr = \frac{2(-1)^{n+1}}{\pi n}.$$

Thus, we obtain the following solution:

$$u(r,t) = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}\sin(\pi n t)\sin(\pi n r)}{\pi^2 n^2 r}.$$

□